How a building stands up

Every building does two jobs: it **holds things up**, and it **stays standing when pushed sideways**. A structural engineer designs the path every load takes to the ground. That is what KOR is paid for, and it is the whole of this lesson.

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01 Job one: holding things up
02 Job two: staying up when pushed
03 Where it all ends: the ground
04 The decisions clients feel
05 Words to use
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07 Check yourself

JOB ONE

Holding things up: gravity

Gravity loads come in three kinds, and the engineer adds them up floor by floor, all the way down.

- **Dead load:** the building itself. Concrete is heavy, about 2,400 kg per cubic metre, so in a concrete tower the structure is most of what the structure carries.
- **Live load:** people, furniture, cars. The building code sets it by use; a parkade, a classroom and an apartment are each designed for a different number.
- **Snow:** set by location. It matters most for roofs in the Interior and Alberta.

The **load path** is the route each load takes: slab → beams → columns or walls → foundations → soil. Columns stack straight down where they can.

✓ THE WORD THAT COSTS MONEY: TRANSFER

When a column or wall above can't continue straight down (a tower over an open lobby, or a different grid in the parkade), a very deep beam or a thick **transfer slab** carries it sideways to new supports. Transfers are heavy, slow to build and expensive. Avoiding them is one of the most valuable things a structural engineer does, and the only time to do it cheaply is at concept.

JOB TWO

Staying up when pushed: wind and earthquakes

Wind and earthquakes push a building sideways. In coastal BC, **earthquake usually governs**: the Lower Mainland and Vancouver Island are among the most seismic places in Canada. Wind can govern for a tall, slender tower.

- **The lateral system** is whatever resists the push. In a concrete tower it is the **core**: the concrete walls around the elevators and stairs. In wood frame it is **shear walls**, plywood-sheathed walls tied down to the floor below. In steel it is **braced frames** (diagonal members) or **moment frames** (stiff beam-column joints).
- **Floors are diaphragms:** horizontal plates that collect the sideways force at each level and deliver it to the core or shear walls.
- **Ductility:** buildings are designed to bend and absorb a big earthquake without collapsing, not to come through undamaged. The code's main goal is that people get out alive. For buildings over 30 m in the highest seismic categories, the 2020 National Building Code adds a requirement to limit damage as well.

KOR'S OWN LINE ON WOOD

KOR tells clients that the Building Code prohibits shear-wall transfers in wood-frame buildings over four storeys, so KOR should be brought in at schematic design. Five- and six-storey wood lives or dies on walls that stack.

WHERE IT ENDS

The ground

Every load ends in the soil. The **geotechnical engineer** reports what the ground can carry and how it will behave in an earthquake; the structural engineer designs the foundations to suit, and tells the geotechnical engineer the loads, the settlement the building can tolerate and the seismic requirements.

- **Spread footings:** pads under columns and walls, on good ground.
- **Raft:** one thick slab under the whole building, to spread a heavy tower's weight.
- **Piles:** columns driven or drilled down to firmer ground.
- **Soft or liquefiable soils** (Richmond and much of the Fraser delta) mean ground improvement or piles, and real money. Ask early whether there is a geotechnical report.

WHY THE CLIENT CARES

The decisions clients feel

Clients don't buy calculations. They feel a handful of early decisions in cost, schedule and how many units fit, and the structural engineer drives all of them.

- **Core size and location.** EGBC's guideline for tall concrete buildings says the structural engineer should agree core size, location and openings, structural irregularities and column positions with the architect early.
- **The column grid,** and whether it works for both the suites above and the parking below. If not: transfers.
- **Slab thickness.** It sets floor-to-floor height, which sets how many storeys fit under a height limit. An inch saved per floor adds up over thirty floors.
- **Wood, concrete or hybrid.** Five- and six-storey wood frame, and encapsulated mass timber up to twelve storeys, are now real options in BC. Each changes the structure, the schedule and the cost.

✓ WHAT KOR BUILDS

Mostly concrete residential towers and mid-rise wood or concrete multifamily: One Burrard Place, 54 storeys over 8 parking levels (Reliance, Jim Pattison Group); Harwood, 32 storeys (Intracorp); Mod at 1150 Cook, Victoria, 15 storeys; N3 in Calgary, 17 storeys with no car parking.

VOCABULARY

Words to use

- **Load path:** how a load gets to the ground.
- **Gravity system / lateral system:** what holds the building up / what keeps it standing sideways.
- **Core:** the concrete walls around elevators and stairs; a tower's backbone.
- **Shear wall:** a wall that resists sideways force.
- **Diaphragm:** a floor acting as a horizontal plate.
- **Transfer:** a beam or slab that moves a load sideways because the support below isn't there.
- **PT slab (post-tensioned):** a concrete slab with steel tendons stretched tight after it hardens, so it can be thinner and span further. Standard in Vancouver towers.
- **Drift:** how far a floor moves sideways in wind or an earthquake.
- **SER (Structural Engineer of Record):** the engineer responsible for the structure as a whole.
- **Geotech:** the geotechnical engineer; the soil specialist.

IN THE ROOM

What to ask, and what not to say

Five questions for an architect at concept. Each shows you know where the cost is, and each is a reason to bring KOR in now.

1. Where is the core, and is it fixed yet?
2. Does the tower grid line up with the parking below, or are we looking at transfers?
3. What's your height limit and floor-to-floor? How many storeys are you trying to fit?

4. Is there a geotechnical report yet? Anything soft or wet on the site?
5. Concrete, wood or a hybrid, and has a five- or six-storey wood option been tested?

> ✕ DON'T
>
> Don't give sizes, thicknesses or costs, even rough ones. The right answer is "Good question, let me get [engineer] on a call this week." That line also gets you the second meeting.

TWO MINUTES

Check yourself

1. What are the two jobs of a structure?

Carry gravity loads to the ground, and resist sideways loads from wind and earthquakes.

2. In a concrete tower, what resists the earthquake?

Mostly the core: the concrete walls around the elevators and stairs.

3. Why do transfers matter to a developer?

They're heavy, slow and expensive, and they can only be designed out early.

4. Why does slab thickness matter?

It sets floor-to-floor height, so it decides how many storeys fit under a height limit.

5. A site in Richmond. What do you ask first?

Is there a geotechnical report? Soft or liquefiable soil means ground improvement or piles.

> SOURCES, AND THE REVIEW THIS STILL NEEDS
>
> KOR's own portfolio and proposal text; EGBC's Professional Practice Guidelines for tall concrete, Part 3 and geotechnical services; the joint guidelines for encapsulated mass timber. A KOR engineer reads each lesson before it is used with a client.

Concrete, wood, mass timber and steel

Choosing the material is one of the first big decisions on a project, and one of the most expensive to change later. Each material is good at some things and bad at others. Developers pick on **cost, speed, height and carbon**. The structural engineer's job is to tell them early what each choice really costs. This lesson tells you enough to ask the right questions.

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01 Concrete: towers, podiums, parkades
02 Wood frame: the five- and six-storey market
03 Mass timber: up to twelve storeys
04 Steel: long spans and hybrids
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MATERIAL ONE

Concrete: towers, podiums and parkades

Concrete is strong when squeezed and weak when pulled. Steel reinforcing bars (rebar) cast inside it take the pulling. Together they make **reinforced concrete**, which is most of what KOR designs.

- **Good at:** height, fire resistance, sound separation between suites, durability, and stiffness. It does not burn, and it can be cast into almost any shape.
- **Bad at:** weight and speed. It is heavy, so it puts more load on the foundations and attracts more earthquake force. Each floor waits for formwork, rebar, a pour and curing.

- **Where KOR uses it:** residential towers, the podiums under them, and parkades. Below grade, almost everything is concrete, whatever sits on top.

Post-tensioned (PT) slabs are standard in Vancouver towers. Steel tendons run through the slab inside ducts. Once the concrete has gained strength, the tendons are stretched tight and locked off. That squeezes the slab and lets it be thinner and span further. A thinner slab means a lower floor-to-floor height, which can mean more storeys under a height limit.

> HOW PT WORKS ON A KOR JOB
>
> A specialty engineer working for the contractor usually lays out the tendons and anchors. KOR, as structural engineer of record, reviews that work and takes overall responsibility for it, as EGBC's Part 3 guideline describes. KOR's specification also requires the slab's strength to be verified before stressing, and a materials consultant on site full time when suspended PT slabs are poured. One catch for owners: you can't drill or shoot anchors into a PT slab without the engineer's approval, because you might cut a tendon.

> ✓ KOR'S CONCRETE RECORD
>
> The Paradox Hotel at 1161 West Georgia, 63 storeys over nine parking levels, a twisted Arthur Erickson tower for Holborn Group and TA Global; The Standard in Burnaby for Anthem, a 43-storey tower beside a six-storey wood-frame rental building over shared parking; Hudson Place in Victoria for Townline, 28 storeys. KOR's own BD view is that mid-rise concrete is its competitive sweet spot for BC Housing's supportive and transitional housing.

MATERIAL TWO

Wood frame: the five- and six-storey market

Light wood frame is studs, joists and plywood: the way houses are built, scaled up. BC allows it to six storeys, usually over a concrete podium and parkade. That makes it a big market.

- **Why developers want it:** it is usually cheaper and faster to build than concrete, it uses trades that are widely available, and it is light, so foundations are smaller.

- **The usual shape:** four to six storeys of wood on a concrete podium, with concrete parking below. KOR's Monument at 2485 Larch Street, Vancouver, is wood frame and cross-laminated timber on a concrete podium over two parking levels.
- **What changes going from four storeys to five or six:** EGBC's bulletin on mid-rise wood says sideways loads go up, shrinkage adds up, framing gets heavier and the fire requirements get stricter. Structural, fire and envelope engineering all take more effort and more coordination.

There are three structural catches. Each is a reason to bring KOR in at schematic design.

- **Shrinkage.** Wood shrinks as it dries, mostly across the grain: in the plates and joists at every floor. Over six floors it adds up, and the concrete stair, the elevator, the plumbing stacks and the cladding don't shrink with it. EGBC's bulletin gives a planning allowance of about 20 mm (3/4 inch) per floor unless the engineer does a more refined analysis. KOR's specification requires the tie-down rods in shear walls to have a shrinkage take-up device at every storey, with no storeys skipped.
- **Shear walls must stack.** In wood, the walls that resist earthquakes are plywood-sheathed walls tied down to the floor below. They need to sit on top of one another all the way down to the podium.
- **No transfers over four storeys.** KOR tells clients that the Building Code prohibits shear-wall transfers in wood-frame buildings over four storeys. So the shear walls have to be built into the unit plans and the massing from the start. If the architect lays out suites first and calls KOR later, walls may need to move, and suites with them.

> KOR'S OWN LINE ON MID-RISE WOOD
>
> KOR markets a dedicated wood-frame design department, and presents itself as experienced in numerous five- and six-storey wood-frame projects and expert in their special design issues. Examples: Chelsea Marine on SW Marine Drive, two six-storey buildings with 102 affordable units for the Vancouver Affordable Housing Agency, completed in 2022; Salt & Meadow in Tsawwassen, three six-storey buildings for Aquilini Development; Parker House in Coquitlam, two five-storey buildings for Polygon.

> ✓ WHERE WOOD SITS IN KOR'S BD
>
> KOR's 2026 BD view is honest about this: stick-frame wood is a less differentiated service than concrete, and very small wood-frame jobs fall below KOR's engagement threshold. But a six-storey wood first phase can position KOR for the larger concrete phases that follow.

MATERIAL THREE

Mass timber: up to twelve storeys

Mass timber means large solid panels and beams made in a factory: **CLT** (cross-laminated timber, layers of boards glued crosswise into slabs and walls) and **glulam** (glued-laminated beams and columns). BC now allows **encapsulated mass timber** up to twelve storeys.

- **The code line:** in BC, buildings of seven to twelve storeys may be encapsulated mass timber or non-combustible construction. Light wood frame stops at six. "Encapsulated" means the timber is covered, usually in gypsum board, to protect it in a fire.
- **Good at:** speed of erection, light weight, precision, lower embodied carbon, and a look that clients and the public like.
- **Bad at:** cost can be higher, the supply chain is narrower, and it needs much more coordination early, because panels are made in a factory and changes are expensive once ordered.

The catches a developer should know about:

- **The core is usually concrete.** The stair and elevator shafts usually form part of the lateral system. The joint EGBC/AIBC guideline says CLT shear walls are height-limited, so timber cores above grade are unlikely in high-seismic areas like coastal BC. In practice: a concrete core, with timber floors and columns around it.
- **CLT shear walls can't do the whole job when it's tall.** The same guideline says code height limits, and the CSA wood standard's restriction to platform framing, mean CLT shear walls cannot be a code-compliant lateral system in tall encapsulated mass timber buildings.
- **Timber and concrete are built to different tolerances.** The guideline says timber is built to a few millimetres while a twelve-storey concrete core may be out by 25 to 40 mm. Every connection between them has to absorb the difference.
- **Fire and char.** Wood in a fire chars on the surface, and the char protects the wood inside. Where timber is left exposed, the structural engineer must calculate the extra char depth and size members for it. Exposed timber means bigger members.
- **Shrinkage and smoke.** The structural engineer calculates mass timber shrinkage too. Most seven-to-twelve storey buildings need smoke control, so timber shafts need careful airtight detailing.

WHERE KOR STANDS IN MASS TIMBER

KOR has designed long-span glulam on the Port Coquitlam Community Recreation Centre (glulam beams on wood V-columns, with steel trusses), and CLT levels at Monument on Larch Street. KOR's own BD view is that in mass-timber hybrid work it competes with Equilibrium, Fast + Epp and ASPECT. Jericho Lands Phase 1 (MST Development) plans about 35% mass timber, and KOR's 2026 BD report saw it as a direct fit.

MATERIAL FOUR

Steel: long spans, industrial and hybrids

Steel is strong in both pulling and squeezing, and much lighter for its strength than concrete. It is made in a shop and bolted or welded on site.

- **Good at:** long spans (arenas, warehouses, gyms, roofs), speed of erection, and being bent in an earthquake without breaking. It is also easy to modify later.
- **Bad at:** fire, because it loses strength when hot and needs fire protection in most buildings. It can be costly for residential, it transmits sound and vibration more than concrete, and it needs corrosion protection where exposed.
- **Where KOR uses it:** single-storey warehouses with steel roofs, arena and recreation roofs, and hybrids. KOR designed nine tilt-up warehouses with steel roofs at Golden Ears North Business Park in Pitt Meadows for Onni Group. Ross Place in Victoria, eight storeys of seniors rental, used load-bearing steel studs, open-web steel joists, steel beams and concrete. At Langford Central, Tower B is steel modular above Level 4 with concrete elsewhere.
- **Who designs what:** steel connections and open-web joists are often designed by a specialty engineer working for the fabricator. KOR's steel specifications set the loads and the rules; the fabricator's engineer designs to them.

✓ HYBRIDS ARE NORMAL

Most real buildings mix materials. Concrete parking under wood frame. A concrete core with timber floors. A concrete arena with a steel roof. The skill is in the joints between them, where tolerances, shrinkage and fire ratings all meet. That is where an experienced structural engineer earns the fee.

THE NEW CLIENT TOPIC

Embodied carbon

Embodied carbon is the greenhouse gas released making and building the materials, before anyone moves in. Clients ask about it more each year, and some cities now require it to be counted.

- **The structure is most of it.** EGBC's practice advisory says that under typical structure-and-shell reporting scopes, structural materials are roughly two-thirds of a building's embodied carbon. That puts the structural engineer at the centre of the conversation.
- **It is being regulated.** In jurisdictions like the City of Vancouver that regulate building greenhouse gas emissions, the embodied carbon of the materials must be calculated and reported.
- **Concrete can be cut too.** A building doesn't have to switch to timber to lower carbon. The advisory describes setting carbon targets for concrete mixes, allowing lower-carbon cements and supplementary materials, and using later-age strengths where the schedule allows.
- **Substitutions need the engineer.** EGBC's tall concrete guideline says the structural engineer must assess any material substitution made to lower carbon for structural performance, including whether it is covered by code.
- **Mandates point to timber.** KOR's 2026 BD report reads net-zero mandates on institutional housing, such as UBC's Lower Mall, as a pointer toward mass timber and hybrid concrete.

> ✕ DON'T
>
> Don't promise a carbon number or a percentage saving. An estimate comes from a quantity take-off and validated carbon data, and it is the engineer's job. Say instead: "We can give you an early carbon estimate alongside the structural options."

VOCABULARY

Words to use

- **Reinforced concrete:** concrete with rebar inside to take the pulling.

- **PT (post-tensioned) slab:** a slab with tendons stretched after the concrete hardens, so it can be thinner and span further.
- **Podium:** the concrete base storeys that a tower or wood building sits on.
- **Light wood frame / stick frame:** studs, joists and plywood; allowed to six storeys in BC.
- **Shrinkage:** wood getting smaller as it dries; it adds up over floors.
- **Tie-down / hold-down:** the rods or straps that stop a wood shear wall lifting off in an earthquake.
- **Mass timber:** factory-made solid wood panels and beams.
- **CLT:** cross-laminated timber; boards glued in crossing layers into panels.
- **Glulam:** glued-laminated timber beams and columns.
- **Encapsulated mass timber (EMTC):** mass timber covered for fire protection; allowed to twelve storeys in BC.
- **Char:** the burnt layer that protects wood in a fire.
- **Embodied carbon:** the carbon released making and building the materials.
- **Hybrid:** a building that mixes structural materials.

IN THE ROOM

What to ask, and what not to say

Six questions for a developer about material choice. Each shows you know where the risk is, and each is a reason to bring KOR in now.

1. What's driving the material choice: cost, schedule, height, carbon, or the lender?
2. How many storeys? If it's five or six in wood, are the shear walls stacking in the unit plans yet?
3. What's under the wood: how many podium and parking levels, and does the grid line up?
4. Have you looked at mass timber or a hybrid? Who is your timber supplier, and when do panels need to be ordered?
5. Do you have carbon targets or reporting requirements from the city or a funder?
6. Is there a fire engineer and an envelope engineer on the team yet?

✕ DON'T

Don't say one material is cheaper, faster or greener for their project, and don't quote member sizes, slab thicknesses, shrinkage numbers or costs per square foot. The answer depends on the site, the height and the market. The right answer is "That's exactly the comparison we run at concept. Let me get [engineer] on a call this week."

TWO MINUTES

Check yourself

1. **Why do Vancouver towers use post-tensioned slabs?**

   Tensioned tendons let the slab be thinner and span further, which lowers floor-to-floor height and can fit more storeys under a height limit.

2. **Why does KOR want to be involved at schematic design on five- and six-storey wood?**

   The code prohibits shear-wall transfers in wood-frame buildings over four storeys, so the shear walls have to stack and be built into the unit plans from the start.

3. **What is the shrinkage problem in wood frame?**

   Wood shrinks at every floor and it adds up over six storeys, while concrete stairs, elevators, plumbing and cladding don't. Everything has to be detailed to allow for it.

4. **In a twelve-storey mass timber building in Vancouver, what is the core usually made of, and why?**

   Concrete. The core resists earthquakes, and CLT shear walls are height-limited, so timber cores above grade are unlikely in high-seismic areas.

5. **A client says they want "low carbon". What do you ask?**

   Whether they have a target or a reporting requirement, from the city or a funder. Then offer an early carbon estimate alongside the structural options, without promising a number.

SOURCES, AND THE REVIEW THIS STILL NEEDS

KOR's own portfolio, proposal and specification text, and KOR's 2026 BD reports; EGBC's Professional Practice Guidelines for Part 3 buildings and for tall concrete buildings; the Joint Professional Practice Guidelines for encapsulated mass timber construction up to 12 storeys; the Technical and Practice Bulletin on 5 and 6 storey wood frame residential buildings; and EGBC's Practice Advisory on embodied carbon for structural engineers. A KOR engineer reads each lesson before it is used with a client.

The rules: building codes in BC and beyond

Every building KOR designs answers to a **building code**, and every one gets built only after a city issues a **permit**. Knowing which code applies, who signs what, and when the rules change is how you sound like you belong in the room, and how you spot the next client who needs help.

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THE RULEBOOKS

Which code applies

There is one national model code, and each province adopts its own version of it. The question is never "what does the code say" but "which code, and which edition".

- **National Building Code of Canada (NBC):** the model code. It has no force on its own; provinces adopt it, usually with changes. BC's current code is based on NBC 2020.
- **BC Building Code (BCBC):** BC's version, based on the NBC. Outside Vancouver, the code in force is the BCBC 2024.

- **Vancouver Building By-law (VBBL):** the City of Vancouver writes its own code under the Vancouver Charter. Inside the city it applies *instead of* the BCBC. The current edition is VBBL 2025. It differs in real ways: it has its own Letter of Assurance forms, under the VBBL any retaining wall over 1.2 m high counts as a building and needs an engineer, and some BCBC permissions for exposed mass timber in suites do not apply in Vancouver.
- **Not every address is a city:** UBC is its own jurisdiction, separate from Vancouver, and uses the provincial forms.
- **Alberta:** uses the Alberta edition of the National Building Code. The signed forms are different too: KOR provides Schedules A-2 and B-2 at permit and Schedule C-2 at completion, and stamps Alberta drawings with its APEGA Permit to Practice number.
- **California:** uses the California Building Code, with its own process. The owner must hire a **special inspector** the code requires; KOR, as engineer of record, confirms one is engaged and makes the structural observations the city requires. Tall towers there are often designed by **performance-based design** and go through city peer review and plan check.

> KOR'S OWN WORDING
>
> KOR's current fee template provides the Schedule B under the 2025 Vancouver Building By-law for City of Vancouver projects, or under the 2024 BC Building Code to the local authority elsewhere in BC. If you can say that sentence, you already know the most important split.

BIG OR SMALL

Part 3 and Part 9 buildings

The BC code sorts buildings into two families. Which family a project falls in decides whether a structural engineer is required, and how much of the building they sign for.

- **Part 9:** small buildings. Three storeys or less, a building area of no more than 600 square metres, and an ordinary use such as housing, offices or shops. Houses, small walk-ups, small commercial. Part 9 has "prescriptive" rules: a builder can follow tables without an engineer.
- **Part 3:** everything bigger, plus some uses no matter how small, such as places of assembly and care facilities. Every KOR tower and every mid-rise is Part 3.

- **Part 4** is the chapter on structural design. On a Part 3 building, the structure is designed to Part 4 by an engineer, always. On a Part 9 building, an engineer is only required for the pieces that fall outside the tables, such as a long-span beam or a deep foundation.
- **A change of use can flip it:** turning a Part 9 building into a Part 3 one (a shop into a daycare, say) usually needs Letters of Assurance and can trigger upgrades to the existing building. Settle that with the city early.

> ✓ WHY A SALESPERSON CARES
>
> Part 3 means a structural engineer of record on every project, from concept to occupancy. Part 9 means maybe a few beams. When a lead looks small, ask how many storeys and how many square metres. That tells you whether it is KOR's kind of work.

PERMITS

The permit, and who signs what

The **authority having jurisdiction (AHJ)** is the city or district building department that issues the building permit. It does not check every calculation. It relies on registered professionals signing that they did.

- **Letters of Assurance:** standard signed forms that tell the AHJ the professionals' work is coordinated, that the design substantially complies with the code, and that field reviews will be and have been done. Vancouver introduced them in 1990 and the BC code in 1992.
- **Schedule A:** signed by the owner and the Coordinating Registered Professional. The city will not issue a permit without it.
- **Schedule B:** one per discipline. The structural engineer of record signs the structural one, listing the code items they will design and review on site. The structural items include the structure itself, deep foundations, shop drawing review, post-tensioned concrete, and independent review of the design.
- **Schedule C-B:** signed at the end, confirming the field reviews were done. It goes in before occupancy.
- **The Coordinating Registered Professional (CRP):** the prime consultant who coordinates every professional's design and field reviews and is the contact between the city, the owner and the team. On most building projects the CRP is the **architect**. On design-build, it is usually someone working for the design-build contractor.

- **Struct.Eng.:** EGBC's designation for an engineer approved to be the structural engineer of record on Part 4 buildings. Some cities accept only a Struct.Eng. on certain buildings.
- **Independent review:** EGBC requires a second, uninvolved structural engineer to review the design before it is issued for construction. Some cities want it before the permit.

> ✓ THE ARCHITECT HIRES US, SO THE ARCHITECT MATTERS
>
> Because the architect is usually the CRP and prime consultant, KOR is usually their sub-consultant. The architect picks, or strongly influences, who does the structure. That is why so much of BD is architect relationships.

EARTHQUAKES

Seismic rules, and what changed

In coastal BC the earthquake usually decides the structure. The code's first goal is that the building does not collapse and people get out. It is not designed to come through undamaged.

- **The design earthquake** is a level of ground shaking, not a magnitude. KOR's own seismic memo for clients (SOCO Towers) describes it as shaking with a 1-in-2,450-year return period, and says the design aims to prevent collapse and limit damage.
- **Site class:** how the ground under the building changes the shaking. Soft ground generally shakes harder than rock, so the same building can need more structure on one site than another. The geotechnical report decides it.
- **The hazard maps were updated.** NBC 2020 brought in Canada's newer seismic hazard model, and BCBC 2024 is based on NBC 2020. In parts of BC that means higher seismic demands than before. Which sites and by how much is an engineering question; do not guess.
- **Damage limitation for tall buildings:** NBC 2020 adds a new requirement for ordinary buildings over 30 m in the highest seismic category. The parts of the frame that are not the main earthquake system (in a concrete tower, typically the columns and slabs outside the core) must stay undamaged in a smaller, more frequent earthquake. Ask the engineer whether a given tower falls under it.
- **Gravity frames:** the 2018 BC code and 2019 Vancouver by-law already added major seismic requirements for the columns and slabs of concrete buildings, which matter most for tall towers in the Lower Mainland and on Vancouver Island.

- **Important buildings:** under NBC 2020, institutional buildings in moderate and high seismic zones are also designed to stay usable after a smaller earthquake, on top of the life-safety design. ACEC-BC's fee guidelines add half a percent to the structural base fee for this.
- **Very tall or unusual towers** may be designed using advanced computer analysis. When that analysis is used to justify a concrete building, the standard requires an independent peer review panel.

> WHAT KOR BRINGS TO IT
>
> KOR runs 3D seismic analysis in ETABS and nonlinear analysis in PERFORM3D, and presents Jeremy Atkinson, licensed in BC and California, as its adviser on challenging projects using performance-based design and peer review by seismic experts.

TIMBER

Wood: five, six, and twelve storeys

- **Five- and six-storey wood frame** came into the BC code with the 2009 mid-rise provisions. EGBC's bulletin says going from four storeys to five or six raises the lateral loads, the wood shrinkage and the fire requirements, and needs more coordination, so engineering effort goes up.
- **KOR's line:** the code prohibits shear wall transfers in wood-frame buildings over four storeys, so shear walls have to be built into the unit plans. Bring KOR in at schematic design.
- **A common mix:** concrete at the bottom, wood above. KOR's Cambie Christian Centre in Vancouver is two six-storey buildings with two concrete and four wood-frame levels.
- **Encapsulated mass timber** (large solid timber panels and beams, covered for fire) up to **12 storeys** for residential and office use was added to the BC code on December 12, 2019, and to the Vancouver by-law on July 1, 2020. Most of these buildings are "high buildings" with extra fire and life-safety requirements. The permitted height has been under review since, so check the current limit with the engineer.
- **Mass timber still often has a concrete core.** The code limits timber shear walls to 20 m or 30 m depending on site seismicity, so taller timber buildings usually rely on a concrete core or steel bracing.

WHY IT SELLS

Code changes are BD openings

Every new code edition leaves developers asking two questions: does it apply to my project, and what will it cost me? A firm that answers clearly, early, earns trust and often the job.

- **Timing decides the code.** In general the code in force when the permit is applied for is the one the building is designed to, and a new edition usually comes with a transition period. The exact cut-off for any project is a question for the engineer and the city, not for you.
- **Presales put pressure on the permit.** In BC, developers can start presales at third reading of the rezoning, but then have about 12 months to get a building permit. The structural permit drawings have to be done inside that window.
- **Old designs on the shelf:** a project designed to an earlier code and shelved may need a redesign if it misses the cut-off. That is a review KOR can offer.
- **Higher seismic demands** can mean more concrete, more reinforcing, or a different system. A developer comparing pro formas needs an engineer to say what that means for their building.
- **New permissions open new options:** taller wood, mass timber and mixed systems. Clients want to know whether the option is real for their site before they pay to design it.

> ✕ DON'T
>
> Don't tell a client their project is "under the old code" or "won't be affected". Don't estimate the cost of a code change. Say: "That's exactly the question to put to our engineer. Can I set up a call?"

VOCABULARY

Words to use

- **NBC / BCBC / VBBL:** national model code / BC Building Code / Vancouver Building By-law.
- **AHJ (authority having jurisdiction):** the building department that issues the permit.
- **Part 3 / Part 9:** large or special-use buildings / small, simple buildings.

- **Part 4:** the structural design chapter of the code.
- **Letters of Assurance (Schedules A, B, C-B):** the signed forms that tell the city the professionals take responsibility.
- **CRP (Coordinating Registered Professional):** the prime consultant coordinating the team; usually the architect.
- **SER / Struct.Eng.:** structural engineer of record / EGBC's designation for engineers approved to take that role.
- **Site class:** how the soil changes the earthquake shaking.
- **Independent review / peer review:** a second engineer checking the design.
- **Encapsulated mass timber:** solid timber structure covered for fire protection.

IN THE ROOM

What to ask, and what not to say

Five questions for a developer about code timing. Each one shows you know the rules matter, and each is a reason to get KOR on the phone.

1. Which city is the permit in, and is it Vancouver, so the Vancouver by-law?
2. When do you expect to apply for the building permit? Has the structural engineer confirmed which code edition that falls under?
3. Are you presales-driven? When was third reading, and what's your permit deadline?
4. Was any of this design done to an earlier code? Has anyone checked what the new seismic rules mean for it?
5. Who is the architect, and are they acting as coordinating registered professional?

✕ DON'T

Don't quote code clauses, seismic values, height limits for a specific site or the cost of a code change. Don't say whether a project needs a structural engineer if it is borderline Part 9. Every one of those is the engineer's call, and saying "let me get [engineer] on that" is the right answer.

TWO MINUTES

Check yourself

1. A project is in the City of Vancouver. Which code applies?

The Vancouver Building By-law, not the BC Building Code. Vancouver writes its own.

2. What makes a building Part 9 rather than Part 3?

Three storeys or less, no more than 600 square metres of building area, and an ordinary use. Anything bigger, or an assembly or care use, is Part 3.

3. Who usually signs Schedule A, and why does it matter to BD?

The owner and the Coordinating Registered Professional, usually the architect. The architect is the prime consultant, so they usually decide who does the structure.

4. What did NBC 2020 add for tall buildings in high seismic areas?

Damage limitation: for ordinary buildings over 30 m in the highest seismic category, the frame outside the main earthquake system must stay undamaged in a smaller, more frequent earthquake.

5. A developer asks if the new code will raise their costs. What do you say?

"Good question. It depends on the site and when the permit goes in. Let me set up a call with our engineer." Never estimate it yourself.

SOURCES, AND THE REVIEW THIS STILL NEEDS

KOR's own fee templates, field review procedures, specifications and portfolio text; EGBC's Professional Practice Guidelines for Part 3 structural services, tall concrete buildings, Part 9 structural services and geotechnical services; the Joint Guidelines for Letters of Assurance on Part 9 buildings and for encapsulated mass timber; the EGBC bulletin on 5- and 6-storey wood frame; the provincial Guide to the Letters of Assurance; ACEC-BC's budget guidelines; BC's real estate development marketing policy statements. Code editions and transition dates change: a KOR engineer reads each lesson before it is used with a client.

What KOR actually delivers, from sketch to occupancy

A client sees a drawing set and a signature. Behind them sit years of work: a design that gets the structure into the building, a permit set the engineer **seals**, review of what the trades build from it, and site visits until the engineer can sign that the finished building **substantially conforms**. This lesson walks the job from start to finish, and shows where fee disputes come from.

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THE TIMELINE

The stages, and what KOR does in each

EGBC's guideline for structural services on Part 3 buildings (the larger buildings KOR works on) lists the basic services in order: conceptual design, design development, permitting, contract documents, tendering, and construction, including review of submittals and field reviews. Each stage asks something different of the engineer.

- **Concept and schematic design:** choosing the system: core, grid, slab, wood or concrete, the Lesson 1 decisions, made while they are cheap. KOR commits a project engineer and a senior partner from the first concepts to the end of construction. Once a scheme is chosen, KOR does a risk assessment and a senior engineer from outside the team reviews the concept.
- **Design development:** the heavy lifting. The building is modelled and analysed, members are sized, and the structure is coordinated with the architect and the mechanical and electrical engineers. On one 2021 KOR fixed fee, design development to building permit carried $47,000 of a $63,000 design fee; schematic design to development permit was $6,000.
- **Building permit:** the engineer seals the structural drawings submitted for permit after checking that they substantially comply with the building code. EGBC requires a permit set to be marked *Issued for Permit Purposes Only*, and the engineer may also mark it Not for Construction.
- **Contract documents (issued for construction):** the drawings and specifications the contractor builds from. These are the documents that get the final independent review (below).
- **Tender:** documents bidders will rely on must be sealed and marked *For Tender Only*. Depending on the contract, the engineer answers bidders' questions and issues addenda; one client's scope on a 2018 project also asked the structural engineer to prequalify contractors and write a tender recommendation.
- **Construction:** shop drawing review, answers to contractor questions (RFIs), site instructions, field reviews, and finally the letter that lets the building be occupied.

> WHY THE PERMIT DATE MATTERS TO A DEVELOPER
>
> In BC a developer can start presales at third reading of the rezoning, but must then get a building permit within about twelve months. The permit drawing set, structural included, has to be produced in that window to keep the presales alive. That is often the real deadline behind a client's urgency.

THE DELIVERABLES

What's in the package

- **Drawings:** plans, sections and details of every structural element, sealed by the engineer. They are the legal record of the design.
- **Specifications:** the written rules for materials and workmanship. KOR keeps templates for formwork, reinforcement, cast-in-place concrete, testing, structural steel, metal decking, light steel framing and glulam.

- **Calculations:** dated, legible, kept in the project file with the input and output of any computer analysis. The client rarely sees them; they are what the firm defends its design with.
- **The structural model:** a computer model of the whole building used to analyse it for gravity, wind and earthquake. KOR uses ETABS and PERFORM3D for 3D seismic analysis, RAM Concept and SAFE for slab design, and Revit and AutoCAD for drawings.

THE SIGNATURES

Letters of Assurance: what the signature means

Letters of Assurance are mandatory legal documents under the BC Building Code (and the Vancouver Building By-law in Vancouver). They name the professionals responsible for each part of the building and record what each one promises. Construction cannot start without the full set.

- **Schedule A:** names the Coordinating Registered Professional, the one responsible for coordinating the design and field reviews of all the professionals on the project.
- **Schedule B (at building permit):** one per discipline. The structural engineer confirms the design substantially complies with the code (construction safety excepted) and **commits to field review** the items listed on it. It is a promise about the design and a promise to come to site.
- **Schedule C-B (at the end):** "Assurance of Professional Field Review and Compliance". Filed after completion and before occupancy, it says the engineer has done the field reviews and, based on them, the finished work substantially complies in all material respects with the code and the permit drawings.
- **No conditions, no gaps:** authorities should reject letters marked interim or partial, or that list deficiencies, and letters must not be issued for any portion that has code deficiencies. In Alberta the same letters are Schedules A-2, B-2 and C-2; KOR's standard scope includes them on both sides of the border.

✕ LETTERS ARE NOT LEVERAGE

EGBC says an engineer must not withhold sealing a completed document to force payment or settle a contract dispute, and calls it unprofessional. Letters of Assurance must not be used for any purpose other than the one they were written for. If you come across older contract wording that ties letters to payment, don't quote it to a client; that practice is under review.

WHO ANSWERS FOR WHAT

Who else signs: the SER and supporting engineers

KOR is usually the **Structural Engineer of Record (SER)**: responsible for the structural integrity of the primary structure and, on Schedule B, for every structural item. But not every piece is designed by KOR.

- **Supporting Registered Professionals** (often called specialty engineers) design specific components. Common examples: guardrails, trusses, cladding and glazing, seismic restraint of mechanical and electrical equipment, and prefabricated systems. They can be hired by the owner, the SER, the contractor or a sub-trade.
- **PT tendons:** on post-tensioned slabs, a specialty engineer working for the contractor usually lays out the tendons and anchors. The SER still reviews that work and takes overall responsibility for it on the Schedule B.
- **How they report:** each supporting engineer seals their own drawings, agrees their scope with the SER, field reviews their own elements and gives the SER sealed Schedules **S-B** and **S-C**. The SER relies on those to sign its own letters.
- **The trap:** KOR's own procedures warn that if a specialty shop drawing arrives without the specialty engineer's seal, the SER may become responsible for the design shown on it. That is why KOR's drawings and specifications state which elements need a specialty engineer.

CHECKING THE CHECKER

A second pair of eyes: independent review

EGBC requires a documented **independent review** of the structural design before documents are issued for construction, by a qualified engineer who was not involved in the design. It shows on the Schedule B as item 2.5.

- **Type 1 and Type 2:** a Type 1 reviewer may work at the same firm; a Type 2 reviewer may not. On tall concrete, the SER decides which by a risk assessment after concept; EGBC recommends Type 2 for irregular or uncommon buildings, such as several towers on a shared podium. "Tall" is about context, not storeys: a ten-storey building among low-rise neighbours can count.

- **Peer review for advanced analysis:** where the design uses non-linear dynamic analysis (a detailed computer simulation of how the building moves in an earthquake), the SER must write a Basis of Design document and have a peer review completed before issue for construction. In the US, KOR prices performance-based design and peer review as an option that applies only if the owner chooses a solution that needs it.

> ✓ SELLING IT
>
> KOR's method has drawings independently reviewed by another team member before they are issued for construction, after tender feedback is in. That is a fair answer to "how do I know the design is right?"

ON SITE

During construction

KOR's standard scope includes the field reviews the code requires, structural shop drawing review, review of testing agency reports, RFI responses and Structural Site Instructions.

- **Shop drawings:** the trades' own drawings (rebar, steel, PT, precast) showing exactly how they will build it. The engineer reviews them for general conformance with the design, not to check dimensions, quantities or construction safety, and confirms specialty items carry the specialty engineer's seal.
- **Field reviews:** site visits, by the engineer or someone under their direct supervision, to judge whether the work substantially complies in all material respects with the design. They can include the fabrication shop. Contractors ask for one by noon the day before; KOR uploads each report the same day, listing observations and deficiencies, to the contractor and the prime consultant, and to the owner when the owner hired KOR directly.
- **How many:** EGBC is clear that the timing and number are the engineer's call, set by the risk, complexity and the contractor's experience. An engineer may estimate the number for a fee, but may not agree in a contract to cap it in advance.
- **Temporary works belong to the contractor:** formwork, shoring, reshoring and falsework are designed by an engineer the contractor hires. KOR's formwork specification says so in terms. The contractor is responsible for means, methods, sequence and site safety.

✕ FIELD REVIEW IS NOT INSPECTION

A field review is not supervision of construction, not a site safety inspection, and not a guarantee that every deficiency will be found. The contractor supervises the work. The engineer reports what does not conform; it does not tell the contractor how to fix it. One exception: if the engineer sees a safety hazard, it must tell whoever controls the site, and if people are in imminent danger, act to stop the work. Never describe KOR as "inspecting" or "supervising" the build.

KOR'S FEE TERMS, AND THE RULE THAT WINS

KOR's current fixed-fee template usually states a number of field reviews with extra visits billed per visit. EGBC allows an estimate but not a limit on the engineer's discretion, so that wording is under review. Say "the fee assumes about N reviews; the engineer decides what the job needs" — never "you get N reviews".

WHY IT MATTERS IN BD

Scope, extras and fee disputes

Almost every fee argument starts with a scope that was never written down clearly. EGBC's guidelines say scope, basic and additional services, fees and payment terms should be agreed before work starts.

- **Basic services** are everything above, including checking, independent review and field review. ACEC-BC's fee guideline says these quality steps are part of the fee, not extras.
- **Additional services** are work caused by changes after the agreement: changes to scope, size or design. KOR's templates list revisions after design acceptance, record drawings, site work outside the base building and alternate-bid documents; KOR's procedures add review of the contractor's temporary works, continuous field review and re-review of improper submittals.
- **The classic one:** the architect moves the program. KOR prices that as a fixed additional fee for coordination, new column locations, slab redesign, re-analysis of the core and drawing revisions.
- **How KOR documents it:** additional services only with the client's written authorization, recorded in an addendum or contract change form the client signs, priced as a fixed fee or hourly to a maximum.
- **A low fee doesn't lower the bar:** EGBC says an insufficient fee does not justify services below its guidelines. Cutting the fee means cutting scope, not care.

✓ THE VALUE STORY IN ONE BREATH

"We get the structure into your building at the stage when it's cheap to change, we get you a permit set on your presale timeline, we stand behind it with our seal, and we're on site until we can sign you into occupancy."

VOCABULARY

Words to use

- **IFP / IFT / IFC:** issued for permit, for tender, for construction. Three stamps on the same set, each a different promise.
- **Schedule B / C-B:** the structural letter at permit (design and a commitment to field review) and at completion (field reviews done, the work conforms).
- **SER:** Structural Engineer of Record; KOR on most of its projects.
- **Supporting Registered Professional / specialty engineer:** designs one component and reports to the SER on Schedules S-B and S-C.
- **Shop drawings:** the trades' drawings, reviewed for general conformance.
- **Field review:** the engineer's site visit. Not "inspection".
- **RFI / SSI:** request for information from the contractor; Structural Site Instruction from KOR.
- **Independent review:** a documented check by an engineer who did not do the design. Type 1 same firm, Type 2 another firm.
- **Temporary works:** formwork, shoring, reshoring. The contractor's engineer.
- **Additional services:** work outside the agreed scope, done on a signed change.

IN THE ROOM

What to ask, and what not to say

Five questions before a fee goes out. Each pins down scope, which is what prevents the argument later.

1. When do you need the permit set? Are you presold, or about to be?

2. How is the job being delivered: tendered, construction management, design-build?
3. Who is hiring the specialty engineers: PT, precast, guards, cladding?
4. Is the design settled, or is the program still moving?
5. Do you want us through construction, and is it one phase or several?

> ✕ DON'T
>
> Don't say KOR "inspects", "supervises" or "guarantees" the work. Don't promise a fixed number of site visits, or a letter "once the invoices are paid". Don't offer to "just stamp" someone else's drawings: an engineer may seal only work done by them or under their direct supervision, and EGBC bars charging a fee for sealing alone.

TWO MINUTES

Check yourself

1. **What does a structural engineer promise on Schedule B?**

   That the design substantially complies with the code, and that it will field review the work.

2. **And on Schedule C-B?**

   That the field reviews are done and, based on them, the finished structure substantially complies with the code and the permit drawings. It is needed before occupancy.

3. **A contractor says "your engineer missed that crack on site." What's the answer?**

   Field review is not supervision or a guarantee that every deficiency is found; the contractor supervises the work and is responsible for building to the documents.

4. **Who designs the shoring and formwork?**

   An engineer hired by the contractor. The contractor owns temporary works, means, methods and site safety.

5. **The architect moves the core after design is accepted. What happens to the fee?**

   It is an additional service: priced, written up and signed by the client before the work is done.

SOURCES, AND THE REVIEW THIS STILL NEEDS

EGBC's Guide to the Letters of Assurance; Professional Practice Guidelines for Part 3 structural services, tall concrete buildings, temporary structures, supporting registered professionals and peer review; EGBC's quality management standards for field reviews, independent review and authentication of documents; the Permit to Practice manual; ACEC-BC budget guidelines; KOR's field review and administration procedures, specification templates and fee proposal templates. KOR's fee terms on field reviews and letters are being reviewed against EGBC's current standards. A KOR engineer reads each lesson before it is used with a client.

Fees: how they're built, and why we win or lose

A structural fee is a **promise of effort, phase by phase**, priced against the cost of the structure. You will never set one. But you will be in the room when a developer says "that's high", and you need to know what the number is made of, what the industry says is reasonable, and what KOR's own record says about which fees we win.

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02 The industry benchmark: ACEC-BC
03 Who the client is
04 Phases, field reviews and extras
05 The proposal itself
06 KOR's record: what we win
07 Words to use
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09 Check yourself

THE BASICS

Three ways to charge

Every structural fee is one of three things, or a mix of them. Knowing which one a client is looking at tells you what they are really comparing.

- **Percentage of construction cost:** the fee is a percentage of what the structure costs to build. It is the industry's benchmark method for the main design work. Its weakness is that the contract must say exactly which costs count as "construction cost"; if it doesn't, the argument comes later.

- **Fixed fee (lump sum):** one number for a defined scope. ACEC-BC says it suits a project whose scope and schedule are defined well enough to estimate the effort, and that it gives the client cost certainty. It is usually built from time estimates or from the percentage method. KOR's standard proposal is a fixed fee, split into phases.
- **Hourly (time basis):** billed against a published rate schedule. ACEC-BC recommends it when the scope cannot be defined, or when the consultant doesn't control the time needed. It is also how additional services are normally paid.

> KOR'S HOURLY RATES (EFFECTIVE MAY 1, 2025, CAD)
>
> Managing Principal $275 · Principal $250 · Associate Principal $225 · Structural Engineer (P.Eng.) $200 · designers, EITs, technologists and construction field reviewers $175. For comparison, KOR's schedule effective October 1, 2020 topped out at $250 per hour for managing principals and senior engineers. Rates exist so extras can be priced; they are not how a building fee is usually quoted.

THE BENCHMARK

The industry benchmark: ACEC-BC Budget Guidelines

ACEC-BC (the association of consulting engineering firms in BC) publishes Budget Guidelines, Document 2: Buildings. The current edition is November 2022. It sets out fees it considers normally appropriate for the level of service EGBC's practice guidelines require, and it is meant to help owners set budgets. The figures are **minimums**, not a price list.

The percentage applies to the structure, not the whole building. The structural cost base includes: preloading, ground improvement, shoring, excavation, backfill and underpinning; foundations and piling; concrete with rebar and formwork; structural steel and deck; structural timber, including glulam, mass timber and engineered wood; masonry; light gauge steel; precast; and the supports for architectural, mechanical and electrical items that the engineer must design or check. A developer who quotes our fee as a percentage of total project cost is comparing apples with the orchard.

2022 minimum structural fees, market multiple housing (percent of structural construction cost):

- **Duplex or townhouse on slab on grade:** 5.5%
- **Low-rise, 1–4 storeys over a concrete garage:** 5.0%

- **Mid-rise, 5–6 storeys, non-combustible:** 4.5%
- **High-rise, 7–15 storeys:** 4.0%
- **High-rise, over 15 storeys:** 3.5%

The guideline allows unit repetition to reduce the design part for townhouses and low-rise. For general new buildings it grades by complexity and size; for buildings over 5,000 m^2 the minimum runs from 4.0% (simple) to 7.0% (specialized). Tenant improvements, restoration and seismic upgrades are priced on time or lump sum, not as a percentage.

> ✓ WHY THE PERCENTAGE FALLS AS THE BUILDING GROWS
>
> A 30-storey tower repeats the same floor many times. The structure costs far more, but the design effort doesn't grow in proportion. So the percentage falls even as the dollar fee rises. That is why a tall tower can carry a large fee at a low percentage.

WHO PAYS

Who the client is: owner or architect

- **Architect as prime:** the usual set-up. The owner hires the architect, and the architect hires structural, mechanical and electrical as subconsultants. The ACEC-BC percentages are written for exactly this case.
- **Owner direct:** many developers hire structural themselves. The fee is the same work, but the relationship, the contract and who signs the change orders all move to the owner.
- **Engineer as prime:** when the structural engineer leads the whole team, ACEC-BC says time basis should be used instead of the percentage.

This matters in the room because the person you are talking to may not be the person who pays. An architect can champion KOR and still lose the argument with the developer's development manager over the number.

HOW THE FEE IS SPENT

Phases, field reviews and extras

A fee is not one lump of work. It is billed as the design moves through stages, and the stage split tells the client when the effort falls.

- **ACEC-BC's minimum split** for a percentage fee: schematic design 15%, design development 15%, contract documents 40%, tender 5%, construction-related services 25%.
- **KOR's standard Canadian proposal** has four phases: schematic design to the development permit (DP) application; design development to the building permit (BP) application; working drawings to issued-for-construction (IFC); and construction administration with a stated number of field reviews.
- **Early-weighted work:** the joint guidelines for encapsulated mass timber say fees should give a greater share to the early design stages, because that is where the coordination happens. The same logic applies whenever the hard decisions move earlier.

Construction-phase fees. During construction KOR reviews shop drawings and makes **field reviews**: site visits by the engineer, or someone under their supervision, to check that what is built matches the design. KOR's template states how many reviews are included; extras are billed per visit. In KOR's 2025–2026 template that is $350 per visit for up to two hours, then hourly. An alternative term fixes the construction fee regardless of the actual number of visits.

Additional services and change orders. Work outside the proposal scope is billed on top of the fixed fee, at hourly rates. KOR's change form prices a change either on hourly rates with a maximum upset amount, or as a fixed fee, and it needs the owner's signed authorization. KOR tells clients it has no hidden fees: extras are billed only after the owner approves them by purchase order or change order. The exceptions are wasted site visits, such as a same-day cancellation or a site that isn't ready, which are billed hourly.

> ✕ DON'T PROMISE A SCOPE CHANGE IS "COVERED"
>
> When an architect mentions a redesign, a new tower or an added parking level, don't say "that's in our fee". Whether it is in scope is the project principal's call, made against the proposal wording.

THE DOCUMENT

The proposal itself

- **The proposal confirms, it doesn't explore.** The books on professional services pricing agree that a proposal should write down what has already been discussed. *Pricing Creativity* (Blair Enns) puts it bluntly: tell the client what the price will be before you write the proposal.
- **Talk about the design, not the price.** *The Trusted Advisor Fieldbook* says rates and margins are almost never the real problem; misunderstandings about scope are. Negotiating on total price alone makes the client think you were hiding something.
- **Follow up by phone.** *Million Dollar Consulting Proposals* (Alan Weiss) says to follow up by phone, not email, at the agreed time. The *Marketing Handbook for the Design & Construction* professions says to contact the prospect after submitting, confirm your interest, ask if anything more is needed and find out when the decision will be made.
- **Losing some on price is normal.** Blair Enns: if a firm is not occasionally losing work on price, it is not charging enough; if it needs to win on price, it is not seen as the expert.
- **Count consistently.** The *Shipley Associates Proposal Guide* says to track every proposal as won, lost, no-bid, cancelled, withdrawn, pending or unknown, define the win rate once, and stick to it.

OUR OWN NUMBERS

KOR's record: what we win

KOR's win/loss record was built from the pursuit trackers on the Management share: 701 pursuits, 2020–2026. Two numbers sum it up. **When a client decides, KOR wins 72% of the time.** But a lot of proposals never get a decision.

- **Decided:** 281 won, 109 lost. That is the 72%.
- **Undecided:** 266 more proposals were never answered or closed out.
- **Conversion:** counting everything sent, KOR converts 43%.
- **Size:** the median proposed fee was $90k on wins and $115k on losses and not-won.

Decided win rate by fee band. The bigger the fee, the more often KOR loses:

- **$10–25k:** 94%
- **$25–50k:** 88%
- **$50–100k:** 76%
- **$100–250k:** 69%
- **$250k and up:** 58%

- **Who we lose to:** Glotman Simpson is the competitor named most often in losses, 6 times. 5 of the 6 losses to Glotman Simpson were fees of $250k or more.
- **Why we lose, as recorded:** mostly "awarded to another firm", and a few "fee too high". The tracker rarely says why the other firm won.
- **The inner circle:** Westland, Northland, Cressey, Domus, Kerkhoff and Polygon nearly always hire KOR when they send us work.
- **Where proposals mostly go nowhere:** NSDA, Wesgroup, Strand, Porte and Starlight. KOR prices their work, and most of it is never awarded to us or never closed.

READ THIS AS A PATTERN, NOT A CAUSE

These are correlations. A large fee doesn't lose *because* it is large. Large jobs may draw more competitors, longer procurement and more people with a vote; the record can't separate those. The record shows where KOR wins; it doesn't prove why. And "awarded to another firm" tells us nothing about whether price, relationship or timing decided it.

✓ THREE THINGS THE RECORD SAYS TO DO

Close out every fee. Call at 30 days; have an answer, won or lost, by 90. 266 open proposals are 266 conversations we never finished, and each one is a free chance to learn why. **Raise the go/no-go bar at $250k and up.** That is where we win least and where the strongest competitor lives, so it is where a pursuit should prove it is worth the effort. **Protect the inner circle.** The six clients who nearly always hire KOR are worth more attention than a new name who mostly doesn't answer.

VOCABULARY

Words to use

- **Construction cost (structural):** the cost of the structure itself, which is the base a structural percentage applies to.
- **Fixed fee / lump sum:** one number for a defined scope.
- **Time basis:** hourly, against a rate schedule.
- **Prime consultant:** the firm holding the owner's contract, usually the architect.
- **Subconsultant:** a firm hired by the prime; structural usually is.
- **DP / BP / IFC:** development permit, building permit, issued for construction: the milestones KOR's phases end on.
- **Construction administration (CA):** shop drawing review and field reviews during construction.
- **Field review:** a site visit to check the work matches the design.
- **Additional services:** work outside the proposal scope, billed on top.
- **Upset amount:** the maximum an hourly change can cost without new approval.
- **Go/no-go:** the decision whether to pursue at all.

IN THE ROOM

What to ask, and what not to say

Five questions that show you understand the fee and take the conversation to the work, not the price.

1. Who will hold our contract, you or the architect?
2. When you say the fee is high, high against what: another proposal, a budget line, or a percentage of total cost?
3. Is the other proposal for the same scope: same phases, same number of field reviews?
4. What's still moving in the design, and how do you want changes handled?
5. When will you decide, and may I call you then?

✕ DON'T

Don't quote a fee, a percentage or a discount, and don't agree that a fee is too high. The right answer is "Let me take that back to [principal] and come back to you this week." That keeps the price with the person who owns it, and it gets you another call.

TWO MINUTES

Check yourself

1. **What does an ACEC-BC structural percentage apply to?**

   The cost of the structure: earthwork, shoring, foundations, concrete, steel, timber, masonry, precast and supports for other trades. Not the total project cost.

2. **Why is the minimum percentage lower for a tower over 15 storeys than for a townhouse?**

   Repetition. The structure costs much more, but the design effort doesn't grow in proportion. So the percentage falls while the dollar fee rises.

3. **How is work outside the proposal scope paid?**

   On top of the fixed fee, usually hourly with an upset amount, and only after the owner signs a change order or purchase order.

4. **KOR wins 72% of decided proposals. Why does it convert only 43%?**

   266 proposals were never answered or closed out, so they never count as wins.

5. **Where does KOR's decided win rate drop lowest, and what does the record suggest?**

   Fees of $250k and up, at 58%, which is also where most losses to Glotman Simpson fall. It suggests a higher go/no-go bar there. That is a correlation, not a proven cause.

SOURCES, AND THE REVIEW THIS STILL NEEDS

ACEC-BC Budget Guidelines for Engineering Services, Document 2: Buildings (November 2022); KOR's fee proposal templates, terms and rate schedules (rates effective May 1, 2025 and October 1, 2020); KOR's contract change form and field review procedures; the joint professional practice guidelines for encapsulated mass timber; KOR's win/loss record from the Management share pursuit trackers, 2020–2026; and *Pricing Creativity*, *The Win Without Pitching Manifesto*, *The Trusted Advisor Fieldbook*, *Million Dollar Consulting Proposals*, the *Marketing Handbook for the Design & Construction* professions and the *Shipley Associates Proposal Guide*. A KOR principal reads this lesson before it is used with a client.

Reading a drawing set in five minutes

A drawing set looks like a hundred sheets of noise. It isn't. It is **organised like a book**: an index up front, the rules next, then plans, then the close-ups. Learn the order and a few symbols, and you can find the core, the grid and the parking layout before the coffee arrives.

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01 How a set is organised
02 General notes: the rules of the job
03 Plans, sections, details, and the bubbles that link them
04 Grids: how everyone finds things
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06 Title block, issues and revisions
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THE MAP

How a set is organised

Every consultant issues their own drawings, and each sheet number starts with a letter for the discipline: **A** for the architect, **S** for structural, **M**, **E** and so on for the others. The number after the letter tells you roughly what kind of sheet it is.

- **Architectural (A) sheets:** the building as people will use it. Site plan, floor plans, elevations (the outside faces), building sections, then wall types, details and finishes. The architect's set is usually the biggest and the first to exist.

- **Structural (S) sheets:** only what holds the building up. Grouped in series by the first digit, and each firm sets its own scheme. A typical pattern: the lowest series for **general notes**, then **foundation plans**, then **floor framing or slab plans** level by level, then **sections and details**, then **schedules**.
- **The first sheet** usually carries a drawing list. Read it first: it tells you what is in the set and how far along the set is.

> HOW KOR NUMBERS SOME OF ITS SHEETS
>
> KOR's own drafting decisions give a few fixed points. The **Design Loads table** sits on **S1.01**. Where a slab carries loads that table doesn't cover (ground floors, podiums, roof decks with planters), KOR adds **Design Load Plans**, starting at S1.11 on concrete jobs and S1.21 on wood frame. Shear wall notes are kept in one place (S3.02 is the example given). On towers, a separate **S4.00 column schedule** covers the typical floors. Treat these as KOR habits, not rules every firm follows.

THE RULES

General notes: the rules of the job

The general notes are pages of small text that most people skip. They are the design brief, written down. Two minutes here tell you what kind of building the engineer thinks this is.

- **Code:** which building code and edition the design follows: the BC Building Code, the Vancouver Building By-law, or, in Alberta, the Alberta edition of the National Building Code.
- **Design loads:** live loads by use (suites, corridors, parkade, roof), snow, wind and the seismic parameters. Usually set out as a table.
- **Materials:** concrete strength in MPa, often different for columns, walls and slabs; reinforcing steel grade; and for wood or steel, the grades and products.
- **Foundations:** the soil capacity taken from the geotechnical report, and the report it came from.
- **Shop drawings and specialty engineers:** which parts somebody else designs and seals (trusses, guards, canopies, cladding framing). EGBC's bulletin for five- and six-storey wood frame says the structural drawings should state these requirements.

THE VIEWS

Plans, sections, details, and the bubbles that link them

- **Plan:** looking straight down at one floor, cut at about waist height. Most structural sheets are plans: one per level, or one "typical" plan covering a run of identical floors (the title then reads something like "Levels 4 to 14").
- **Section:** the building sliced vertically, as if cut with a knife. It shows how floors, walls and footings stack.
- **Elevation:** a face of the building or of one wall, seen straight on. Shear wall elevations show openings and reinforcing in a wall from floor to floor.
- **Detail:** a small area blown up large: a slab edge, a wall-to-slab joint, a stair landing. Details carry most of the construction information.

How they reference each other. A plan has small circles on it, usually split by a line. The top half is the **detail or section number**, the bottom half is the **sheet** it is drawn on. A section mark adds an arrow or a cut line showing which way you are looking. "3 over S5.02" means: drawing 3, on sheet S5.02. Follow the bubble, find the drawing; the drawing's title carries the same bubble pointing back.

✓ SLAB CALLOUTS WORTH RECOGNISING

Slab edge: the outline of the floor plate; balconies and cantilevers are where it pushes out past the columns. **Openings:** a box with an X through it is a hole in the slab (shafts, stairs, ducts). **Steps:** a change in slab level, marked with the two top-of-slab elevations or a step symbol, common at balconies, terraces and parkade ramps. **Thickness:** a note or tag on the plan giving slab depth, with thickened areas (drop panels at columns, thicker bands) outlined. A plan crowded with openings, steps and thickness changes is a hard floor to build and to design.

THE COORDINATES

Grids: how everyone finds things

Gridlines are the street address system of a building. Every column, wall and detail is located from them, on every consultant's drawings.

- **Letters one way, numbers the other.** Usually lettered gridlines run one direction and numbered ones the other, each with a bubble at the end. "Column at C-4" is where gridline C crosses gridline 4.
- **The architect's grid is the grid.** Structural drawings use the same lines and names. EGBC's Part 3 guideline expects floor levels, column spacing, wall locations and offsets to be coordinated with the architectural drawings.
- **Half grids** (C.5, or 4.1) are added where something sits between main lines. Lots of them on a plan can mean the layout above and below doesn't line up.
- **Spacing tells you the span.** The dimensions between gridlines are the column spacing: the distance the slab has to span, and the number that has to work for both suites and parking stalls.

THE LISTS

Schedules and reinforcing callouts

A schedule is a list that saves drawing the same thing a hundred times. The plan gives each element a short tag, and the schedule says what the tag means.

- **Column schedule:** each column type (size, concrete strength, vertical bars and ties) listed against the floors it runs through. KOR groups columns by size (PC1, PC2 and so on) instead of tagging each one, and puts the schedule on the plan itself for shorter buildings or complicated levels.
- **Wall schedule:** the same idea for shear walls and core walls: thickness, strength and reinforcing, floor by floor.
- **Beam schedule:** beam tags with size and top and bottom reinforcing.
- **Other tags you'll see:** footings (F1, F2), and on KOR's flat-slab towers, studrails (SR1, SR2): rows of steel studs around a column that stop the slab punching through.

Reinforcing callouts in plain words. Canadian bar sizes are named by approximate diameter in millimetres: 10M, 15M, 20M, 25M, 30M, 35M. A callout gives size, then spacing:

- **"15M @ 300":** 15M bars, spaced 300 mm apart.
- **"15M @ 18":** the same bar at 18 inches. KOR drawings use imperial spacing too; KOR's own minutes give "15M at 18 inches full length" as an example for cantilevered slab top bars.
- **Top / bottom, each way (EW), each face (EF):** where the bars sit. Slabs need bars near the top over supports and near the bottom mid-span.
- **A count instead of a spacing** ("4-25M") means that many bars, as in a beam or a column.

WHY KOR FROZE ITS SCHEDULES

In 2022, schedules that updated themselves in Revit changed a column's concrete strength from 50 MPa to 35 MPa and turned 35M vertical bars into 15M, and back-checks could miss it because nobody had marked it up. KOR changed its template so concrete strengths are fixed values. The point for you: schedules are where small text carries big consequences.

IS THIS CURRENT?

Title block, issues and revisions

The title block is the strip along the right edge or bottom of every sheet. It answers the questions that matter most in a meeting: whose drawing, what project, how current, and is it real.

- **What it holds:** the firm, the project name and address, the sheet title and number, the project number, scale, who drew and checked it, the issue list, and the engineer's seal.
- **The seal:** EGBC places the seal in the title block or lower right corner of each drawing. A sealed drawing carries the engineer's seal, signature and date, and the firm's Permit to Practice number. KOR's BC number is 1000378. On Alberta jobs KOR shows its APEGA permit number (13755) as text in the title block.
- **The issue list:** a small table of dated issues, newest at the top or bottom. Common labels: **Issued for Review** or **for Coordination**, **Issued for Building Permit** (EGBC allows these to be marked Not for Construction), **Issued for Tender**, and **Issued for Construction**, the set a contractor builds from.
- **Revisions:** a change after issue gets a number or letter in the issue list, a **revision cloud** (the scalloped bubble drawn around what changed) and a small triangle with the revision number beside it. EGBC's authentication standard says a revised sealed document shows the revision in the title block, identifies the changes, and is re-sealed and dated.

✓ READ THE STATUS BEFORE THE CONTENT

A set issued for review at schematic design and a set issued for construction look alike from across the table. One is an idea; the other is a commitment, and EGBC requires an independent review of the structural design before it is issued for construction (some authorities want it before permit). The most recent date in the issue list tells you where the project is, and whether the structural engineer is already fixed.

THE OTHER SET

Reading the architect's plan for structure

At a first meeting you will often get only the architect's drawings. They already tell you most of what the structure will be.

- **The core:** elevators and stairs, grouped together. Its size and position decide how the tower resists earthquakes. Off-centre or split cores are harder.
- **Columns:** the small solid squares or rectangles, usually on the grid. Count them; look for an even rhythm. Flip between floors: does each column land on one below?
- **Stacking walls:** in wood frame and concrete mid-rise, the walls between suites often carry the building. Overlay two floor plans (or hold sheets to the window) and see whether they line up.
- **The parkade plan:** stalls and drive aisles want long clear spans; suites above want columns in walls. Where the tower's columns fall in the middle of a drive aisle, someone is going to need a transfer.
- **Sections and elevations:** storeys, floor-to-floor heights and the height limit. Podiums and setbacks are where the structure changes.

HOW KOR MARKS UP AN ARCHITECT'S PLAN

KOR engineers review architectural drawings in Bluebeam and mark the structure in burgundy: the slab edge as a thin outline, walls (cores, stair shafts, shear walls) as heavy closed shapes, columns as small filled rectangles. If you are handed a marked-up PDF from a KOR engineer, burgundy is the structure.

FIVE MINUTES

The five-minute routine

1. **Title block and issue list (30 seconds).** Whose set, which issue, what date. Is there a structural engineer named, and who is it?
2. **Drawing list (30 seconds).** How many sheets, which disciplines. A set with only A sheets is early; a full S series means a structural engineer is already on.
3. **Building section (1 minute).** Storeys above and below grade, podium, floor-to-floor height.
4. **Parkade plan, then a typical floor (2 minutes).** Find the core and the grid on both. Do the columns line up? Any obvious transfer?
5. **General notes, if structural (1 minute).** Code, materials, the geotechnical report reference. On a KOR set, the loads table is on S1.01.

> WHEN A DEVELOPER HANDS YOU A SET AT A FIRST MEETING
>
> Look for four things, and ask about the ones you can't see. **Stage**: what the issue block says. **Who is already on**: consultant names in the title blocks. **The tower-to-parkade fit**: whether the grid survives the trip down. **The geotechnical report**: referenced in the notes or not. Each gap is a reason for the next meeting.

VOCABULARY

Words to use

- **Set:** the full bundle of sheets issued together.
- **General notes:** the code, loads and materials the design is based on.
- **Typical floor:** one plan standing in for a run of identical floors.
- **Grid / gridline:** the lettered and numbered lines everything is located from.
- **Callout / bubble:** the circle that points to a section or detail on another sheet.
- **Schedule:** a list explaining the tags on a plan (columns, walls, beams, footings).
- **Revision cloud:** the scalloped outline around something that changed.
- **IFP / IFC:** issued for permit / issued for construction.

- **Title block:** the strip with the firm, project, sheet number, issues and seal.
- **Shop drawings:** a fabricator's or specialty engineer's drawings of their part, reviewed by the structural engineer for general conformance.

IN THE ROOM

What to ask, and what not to say

Five questions to ask with a set open on the table.

1. Which issue is this, and what's the next milestone: permit, tender or construction?
2. Is a structural engineer already on, or are these architectural only?
3. Is the core fixed, or still moving?
4. Does the parkade grid line up with the tower, or has that not been tested yet?
5. Is there a geotechnical report we can see?

> ✕ DON'T
>
> Don't read a size, a bar or a thickness off a sheet and comment on it ("that slab looks thin", "those columns look big"). You can't tell from the drawing what drove it, and it may be another firm's work in front of their client. Don't mark up a set someone hands you. Say what you noticed as a question, and bring an engineer to answer it.

TWO MINUTES

Check yourself

1. A circle on a plan reads "3" over "S5.02". What is it?

A reference: drawing 3 is on sheet S5.02. Go there to see the section or detail.

2. What does "15M @ 300 T&B EW" mean?

15M bars, 300 mm apart, top and bottom, each way.

3. Where on a sheet do you find whether a set is current?

The title block's issue list: the latest issue, its date and any revision numbers.

4. **On an architect's parkade plan, what are you looking for?**

   The core and the columns, and whether the tower's columns above land on the same grid, or fall in drive aisles and stalls (a transfer).

5. **A set is marked Issued for Building Permit. Can it be built from?**

   Not as such. Permit drawings can be marked Not for Construction; the contractor builds from the Issued for Construction set.

SOURCES, AND THE REVIEW THIS STILL NEEDS

KOR department meeting minutes on drafting standards (sheet numbering, column schedules, studrails, Revit schedules, slab reinforcing); KOR's office procedures (permit numbers, shop drawing review); KOR's specification templates; KOR's markup conventions; EGBC's Professional Practice Guidelines for Part 3 buildings and tall concrete, the five- and six-storey wood frame bulletin, the Permit to Practice manual and the Standard for the Authentication of Documents. Sheet series beyond those named vary by firm and by project. A KOR engineer reads each lesson before it is used with a client.

Talking to architects, developers and contractors

Three kinds of people hire a structural engineer, and each of them is afraid of something different. The first meeting is not a pitch. You **ask about their project until they tell you what worries them**, then bring an engineer back to help with it. This lesson covers how to do that with each of the three.

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02 Three audiences, three fears
03 Advisor, not pitcher
04 Asking good questions
05 First meeting, second meeting, follow-up
06 Three sample openings
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THE CONTRACT

Who actually hires us

EGBC's guideline for structural services on Part 3 buildings names three possible clients for the Structural Engineer of Record: the **owner**, the **coordinating registered professional** (usually the architect), or a **design-build contractor**. KOR's own office procedures say the same.

- **The architect** is KOR's usual client. KOR mostly works as the architect's structural sub, and often gets the job before any RFP exists. This is where most of KOR's work comes from.
- **The developer** pays for everything, and on many projects picks the team or approves the architect's choice. Developers are KOR's references: the proposal reference list names firms such as Cressey, Intracorp, Polygon, Bosa, Onni and Westbank.
- **The design-build contractor** holds the contract when it carries both design and construction. In that case, as EGBC's geotechnical guideline notes, the engineers usually have no contract with the owner at all. They are subconsultants to the builder.

> ✓ WHY THIS MATTERS IN THE ROOM
>
> Before every meeting, work out who would sign KOR's contract on this project, and who has a say without signing anything. Often they are different people. The architect may hold the contract while the developer's preference decides who gets it.

KNOW THE PERSON

Three audiences, three fears

Architects

- **What they care about:** the design surviving. That means the massing, the open ground floor, the unit plans and the floor-to-floor height they promised the developer.
- **What they fear:** a structural engineer who says "no" late. Transfers that eat the budget, walls that don't fit the units, and a redesign after the developer has already seen the scheme.
- **What a structural engineer means to them:** the person who makes the design buildable, or the person who breaks it. KOR's own introductions describe the approach as shaping the structure around the architectural, mechanical and electrical elements instead of making them work around the structure. That is the sentence an architect wants to hear.
- **How they choose:** mostly on trust and past experience. The question is who they have worked with and who makes their life easier. Formally, EGBC and ACEC-BC recommend qualifications-based selection, where the firm is chosen on competence first and the fee is negotiated after.

Developers

- **What they care about:** cost, schedule and how many units fit, and in particular the sellable floor area and how many storeys fit under the height limit.
- **What they fear:** surprises. Costs that jump between concept and tender, a permit that slips, a consultant team that misses their dates. A real-estate finance text in the library notes that a land developer's option period depends on consultants committing to the developer's timetable. Missing it can cost the option and every fee already spent.
- **What a structural engineer means to them:** a large share of hard cost decided early. The early decisions from Lesson 1 (core, grid, slab thickness, wood or concrete) are where structure changes the pro forma.
- **How they choose:** the inner-circle developers usually hire whoever they trusted last time. A developer with land and no team yet is the rare case where KOR can be on the team before the architect is chosen.

Contractors

- **What they care about:** drawings they can build quickly, a price that holds and an engineer who answers questions on site without holding up the pour.
- **What they fear:** a design that looks cheap on paper and is slow to build, slow answers, and change orders they can't recover.
- **What a structural engineer means to them:** on design-build work, a partner whose design they will be held to. Under design-build the contractor carries design responsibility to the owner, and KOR's procedures say the design-build contractor takes on the coordinating professional's role toward the design team.
- **How they choose:** constructability, speed of response and price. The *Marketing Handbook for Design & Construction* notes that contractor-led design-build teams lean toward construction over design and suit cost-driven projects. Design-build also needs a high level of trust and communication inside the team.

> **DESIGN-BUILD AND P3: THE TIMING IS DIFFERENT**
>
> On a design-build-finance-maintain (P3) procurement, KOR's own BD notes say the structural engineer is a sub chosen while the shortlisted consortia form their teams. That leaves roughly four to six weeks before award to get on a team. Know which contractors are shortlisted, and call them during that window.

THE STANCE

Advisor, not pitcher

The business books behind this course agree on one point: **talk about the client's project, not about KOR.**

- **Maister** (*Managing the Professional Service Firm*): talk about the client's situation, not the firm's accomplishments, and treat a prospect as if they were already a client. Be useful from the first meeting.
- **Maister, Green and Galford** (*The Trusted Advisor*): trust has four parts. **Credibility** (what you say), **reliability** (what you do), **intimacy** (whether they feel safe telling you things) and **self-orientation** (whether you seem focused on yourself). The first three add trust. Self-orientation divides it. A pitch is self-orientation at its most visible.
- **Enns** (*Win Without Pitching*): diagnose fully before you prescribe. Gather what you need in early meetings, but don't hand over the diagnosis or the solution before you are hired. For KOR this means offering the conversation with an engineer, not the answer.
- **The sales-engineer pattern:** *The Trusted Advisor Sales Engineer* describes the usual split. The salesperson finds the lead and builds the relationship, then brings in the technical person to show how the problem gets solved. That is your role and the engineer's role.

✕ DON'T

Don't open with a slide deck, the firm's history or a project list. Several of the sales books give the same rule: a first meeting is never a presentation. If you're asked to present, turn it into a conversation by asking questions first.

THE METHOD

Asking good questions: SPIN, adapted to structure

Rackham's SPIN research (*SPIN Selling*, *Major Account Sales Strategy*) sorts the questions that win large sales into four kinds. He found that **Implication** questions, which ask what a problem leads to, are the ones most strongly linked to success when a buyer is still working out what they need. That is exactly the stage KOR wants to be in.

- **Situation:** the facts. "How many storeys? Wood or concrete? Is there parking below? Who's on the team so far?" Ask only what you couldn't find out before the meeting.
- **Problem:** where it hurts. "Does the tower grid work with the parking? Is the height limit tight? Is there a geotechnical report yet?"
- **Implication:** what the problem will cost. "If the grid doesn't line up, does that mean a transfer level? What would that do to the budget, or to the schedule?" "If the shear walls don't stack in the six-storey scheme, what happens to the unit plans?"
- **Need-payoff:** get them to say the benefit out loud. Rackham notes these are generic and reusable: "How would that help?" "Would it be useful if an engineer looked at the grid before you fix it?"

Rackham also stresses that SPIN is not a script. If the client opens with a clear need ("we need to get rid of the transfer"), go straight to need-payoff. Voss (*Never Split the Difference*) adds one technique: phrase questions with **what** and **how** instead of yes/no. "How does this look to you?" beats "Do you like this?". "What's driving the schedule?" beats "Is the schedule tight?". Name what you hear before you ask for anything ("It sounds like the parkade is the worry").

> ✓ THE STRUCTURAL PROBLEMS WORTH LISTENING FOR
>
> Transfers (Lesson 1), a height limit with no room to spare, soft ground, a wood scheme over four storeys where the walls don't stack, and a schedule that depends on the permit. KOR tells clients that the Building Code prohibits shear-wall transfers in wood-frame buildings over four storeys, so shear walls have to be built into the unit plans and massing at schematic design. When an architect describes a five- or six-storey wood scheme, that is your opening.

THE SEQUENCE

First meeting, second meeting, follow-up

The first meeting

- **Prepare lightly:** look at their website and recent projects, check who they've worked with, and look up what is public about the site. Do most of the research in the meeting, by asking. Lencioni's *Getting Naked* makes the same point.
- **Listen more than you talk:** *The Trusted Advisor Fieldbook* says to state your point of view crisply and then spend most of the time listening with curiosity. You've listened well when they turn the conversation back to you.

- **What to offer:** something small and useful that needs an engineer. Good options are an **early concept review** of the grid and core, a **constructability** look at the scheme, or a cost idea such as **designing out a transfer** or **stacking the shear walls** in a wood scheme. KOR's fee templates already allow preliminary design to be billed hourly, capped at the schematic-design fixed fee and folded into that fee if the structure doesn't change much. So an early look doesn't have to be free to be easy to say yes to.
- **End with a date:** Harding's *Rain Making* says that rainmakers end each conversation with an agreed next step, and test the urgency, for example "given what's at stake, can we afford to wait more than two weeks?"

The second meeting: bring the engineer

- **Pick the right engineer** for the building type: a tower person for a tower, a wood person for a six-storey wood scheme.
- **Brief them first:** who's in the room, what problem the client named, and which question you promised an answer to.
- **Let the engineer lead** on the technical side. You run the meeting, keep it on the client's problem and close with the next step.

Follow-up

- **Same week:** a short note that repeats back what you heard and confirms the next step. Repeating it back is reflective listening, which *The Trusted Advisor* lists as a sign of low self-orientation.
- **Keep going:** Ferrazzi (*Never Eat Alone*): follow up, then follow up again. The *Marketing Handbook* quotes rainmakers who found that a relationship takes one to three years to turn into work. A slow start is normal and is no reason to stop.
- **Each follow-up brings something:** a relevant KOR project, a code change, or a question about the stage they're at. "Just checking in" doesn't count.

IN YOUR OWN WORDS

Three sample openings

1 · AN ARCHITECT AT CONCEPT

"Thanks for making time. I'm not here to show you slides. I'd rather hear about the project. Where are you with the massing? Is the core settled? … You mentioned the parking grid is different from the tower above. If that turns into a transfer level, what would it do to the budget you've given the client? … Would it help if one of our engineers sat down with you for an hour on the grid before it's fixed? That's the cheapest time to sort it out."

2 · A DEVELOPER WITH LAND AND NO TEAM YET

"Congratulations on the site. What are you hoping to build there, and what does the timeline look like? … What would have to be true for the numbers to work? … Who are you thinking of for the architect? … A lot of the structure cost gets locked in by decisions in the first few weeks, like wood or concrete, where the core goes and whether the parking lines up. Would it be useful to have an engineer in the room when you and the architect make those calls?"

3 · A CONTRACTOR ON A DESIGN-BUILD

"How are you putting the team together for this one, and when do you need the structural sub confirmed? … What's driving the price, and where do you think the risk is? … What has gone wrong with engineers on past design-builds? … We'd want to design something your crews can build quickly, not something that only looks cheap on paper. How would you like us to work with your estimator before the price goes in?"

Each one asks about the situation first, finds a problem, gets the client to say what it would cost, and offers an engineer as the next step. None of them gives an answer.

THE OBJECTION

"Your fee is too high"

Baker (*Implementing Value Pricing*) argues that a client who says the price is too high usually means they don't yet see the value. So the response is to find out more, not to discount.

- **Ask first:** Weiss (*Million Dollar Consulting*) says to ask why the client feels that way and what would satisfy them, instead of defending each line item. "Too high compared with what? Another proposal, the budget, or last time?"
- **Trade scope, not price:** if the fee has to come down, take out scope of matching value, such as field reviews or phases, rather than cutting the number for the same work. Weiss makes this point in both of his books in the library.
- **Separate price from cost:** the fee is a small number. The cost of a transfer that never got designed out, or a storey that didn't fit, is a big one. Bring the conversation back to that.
- **Competence first:** the *Marketing Handbook* says that within a reasonable range, cost should play a fairly small part in choosing a consultant, and selection should never be a contest of price alone. This is the same principle as the qualifications-based selection EGBC recommends.
- **Leave the number to the engineer:** you can open the conversation, but the principal on the proposal decides any change to scope or fee.

> ✕ DON'T
>
> Don't offer a discount in the room, and don't criticise a competitor's fee. A quick cut tells the client the first number wasn't real.

WHERE THE WORK COMES FROM

The inner circle and referrals

Some clients nearly always hire KOR once KOR proposes: **Westland, Northland, Cressey, Domus, Kerkhoff and Polygon**. They are worth more than any cold lead. The job with them is to stay close enough to be asked to propose every time.

- **Repeat work is the business:** *The Trusted Advisor* notes that trusting relationships bring repeat work from the same client and new work through their referrals. *Clients for Life* calls this attitudinal loyalty: clients who remember the job, call back and recommend you.
- **Hear about projects early:** ask inner-circle clients what's coming next, not only what's live. A site under option is a project KOR can join before the architect is chosen.
- **Follow people when they move:** Harding (*Rain Making*) describes a rainmaker who followed one contact through five companies over twenty years. When a project manager at an inner-circle developer moves to a new firm, that new firm is a warm lead.
- **Ask for referrals rarely and well:** *The Trusted Advisor Fieldbook* advises asking only a couple of times over a good relationship, choosing carefully who to ask, researching the names, and giving the client concrete things to say about you.

VOCABULARY

Words to use

- **Coordinating registered professional (CRP):** the professional, usually the architect, who coordinates the design team and is often KOR's client.
- **Prime consultant:** the firm that holds the owner's contract and hires the subconsultants.
- **Design-build:** one contract for both design and construction. The engineers work for the builder, not the owner.
- **P3 / DBFM:** design-build-finance-maintain. A consortium bids, and the structural engineer is a sub inside it.
- **QBS (qualifications-based selection):** choose the firm on competence, then agree the fee. EGBC and ACEC-BC recommend it.
- **Constructability:** how easily and quickly a design can actually be built.
- **Concept / schematic design:** the earliest design stages, when structural decisions are cheapest to change.
- **Pro forma:** the developer's spreadsheet of costs and revenue that decides whether the project goes ahead.

IN THE ROOM

What to ask, and what not to say

Five questions that work with any of the three. Each is open, and each is about their project.

1. What's the biggest risk on this project right now?
2. What has to be true for the numbers to work?
3. What's already fixed, and what's still open: core, grid, wood or concrete?
4. What went wrong with structure on your last project like this?
5. How would you like us to work with you before anything is decided?

> ✕ DON'T
>
> Don't pitch, and don't give sizes, costs or a diagnosis. Don't promise a fee, a saving or a date. Don't leave without a next step that has a date on it. When a question needs an engineer, say so: "Good question. Let me get [engineer] in front of you this week." That answer also gets you the second meeting.

TWO MINUTES

Check yourself

1. Who can be the Structural Engineer of Record's client?

The owner, the coordinating registered professional (usually the architect), or a design-build contractor.

2. What does an architect fear most from a structural engineer?

A late "no": transfers, walls that don't fit the units, or a redesign after the client has seen the scheme.

3. Which SPIN question is most linked to winning, and what does it sound like here?

The Implication question. "If the grid doesn't line up, what does a transfer level do to your budget and schedule?"

4. A client says the fee is too high. What are your first two moves?

Ask what they're comparing it with and what would satisfy them. If it must come down, take out scope of matching value rather than cutting the price.

5. What do you offer at the end of a first meeting?

A small, useful next step with an engineer, such as an early concept review of the grid and core, with a date agreed.

SOURCES, AND THE REVIEW THIS STILL NEEDS

EGBC's Professional Practice Guidelines for Part 3 structural and geotechnical services; KOR's office procedures, company introductions, fee templates and BD notes; Rackham, *SPIN Selling* and *Major Account Sales Strategy*; Maister, Green and Galford, *The Trusted Advisor*, and its Fieldbook; Maister, *Managing the Professional Service Firm*; Enns, *Win Without Pitching*; Voss, *Never Split the Difference*; Harding, *Rain Making*; the *Marketing Handbook for the Design & Construction Professional*; Baker, *Implementing Value Pricing*; Weiss, *Million Dollar Consulting*. A KOR principal should check the inner-circle list and the fee guidance, and a KOR engineer reads each lesson before it is used with a client.

Thirty questions that make you sound like you know the business

You don't have to know the answers. You have to ask the questions an engineer would ask, **listen to what comes back**, and know when to say "let's get our engineer on a call." This lesson is the questions, the five things never to say, and the whole course on one page.

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01 How to use these questions
02 For an architect at concept
03 For a developer with a site
04 For a contractor or design-builder
05 About the site and the ground
06 About schedule, cost and risk
07 Five things never to say
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BEFORE YOU START

How to use these questions

A good question does two things: it shows you understand where the money and the risk are, and it gives the client a reason to bring KOR in now rather than later.

- **Pick three, not thirty.** A question asked from a checklist backfires. The moment the client thinks "why is he asking me that?", you have lost credibility. Choose the ones that fit the project in front of you.
- **Then listen, and show you listened.** Play back what you heard: "So the core isn't fixed yet. Do I have that right?" Listening is what earns the right to suggest anything.

- **Ask what it would be worth.** "How would it help if we could settle that before you go to the city?" Questions like that let the client explain the value of early structural input in their own words, which is what they will repeat to their partners when you are not in the room.

> ✓ THE LINE THAT ENDS MOST ANSWERS
>
> "That's exactly the kind of thing our engineer should look at. Can I set up a call this week?" It is honest, it gets the second meeting, and it puts a KOR engineer in front of the client at the point where their advice is worth most.

QUESTIONS 1 TO 6

For an architect at concept

Architects choose the structural engineer on most projects, often at the start of schematic design. These questions show you know which early decisions lock in structural cost.

1. **"Where's the core, and is it fixed yet?"**

   Why it works: on a concrete tower the core resists the earthquake. EGBC's tall-building guideline says its size, location and openings should be agreed with the structural engineer early.

   If they say "we're still moving it around": that is the best moment for a KOR engineer to sit in.

2. **"Do the tower columns line up with the parking grid, or are we looking at transfers?"**

   Why it works: transfers are the most expensive structural surprise, and the only cheap time to design them out is now.

   If they say "the parking grid is different": "Let's get our engineer to look at it before it goes further."

3. **"What's your height limit and floor-to-floor? How many storeys are you trying to fit?"**

   Why it works: slab thickness drives floor-to-floor height, so it decides whether the last storey fits. It tells you how tight the envelope is.

4. **"Does anything need to punch through the core at grade, like a parking stair?"**

   Why it works: an opening at the base of the core is exactly where the walls work hardest in an earthquake. EGBC expects the structural engineer to offer other stair layouts that meet the architect's need.

5. **"Concrete, wood or hybrid, and has anyone tested a five- or six-storey wood scheme?"**

Why it works: it shows you know the options. It also opens KOR's line on wood: shear-wall transfers aren't allowed in wood frame over four storeys, so the walls have to be designed into the unit plans from the start.

6. **"If it's mass timber, have you talked to a supplier about panel sizes yet?"**

   Why it works: timber spans differ a lot from concrete. The joint mass-timber guideline says the grid should be set with the structural engineer once the supplier's product sizes and transport limits are known.

   If they say "not yet": "Our engineer can help you set the grid before you commit."

QUESTIONS 7 TO 12

For a developer with a site

Developers care about units, cost, schedule and certainty. Ask about the deal, not the drawings.

7. **"What does the site need to deliver: unit count, height, parking?"**

   Why it works: it starts from their pro forma, not your services. Every structural decision (grid, transfers, slab depth) feeds one of those three numbers.

8. **"Who's your architect, and how far along are they?"**

   Why it works: it tells you where the decision sits. If there is no architect yet, you can make introductions; if there is one, KOR's route in usually goes through them.

9. **"How much parking are you carrying underground?"**

   Why it works: deep parking means excavation, shoring, groundwater and a column grid the tower above has to respect. It is where a lot of the structural cost lives.

10. **"Have you looked at a wood scheme, or is it concrete because that's what you've always done?"**

    Why it works: it gently challenges an assumption, which is what a trusted adviser does. Asked with curiosity, not a pitch.

    If they say "we've never priced wood": "Our engineer can walk you through what changes at five and six storeys."

11. **"What's driving the schedule: a financing date, presales, a permit window?"**

    Why it works: it finds the deadline that actually matters to them, and shows you understand structure is on the critical path to the building permit.

12. **"What went wrong structurally on your last project?"**

Why it works: it brings out real pain (late drawings, slow site reviews, a transfer nobody saw coming), and that pain is the reason they would switch engineers.

If they name a problem: ask "what did that cost you?" before saying anything about KOR.

QUESTIONS 13 TO 18

For a contractor or design-builder

On design-build the contractor is the client, and takes on the job of coordinating the design team. Contractors buy constructability, speed and engineers who answer the phone.

13. "Where are you in the procurement, and when do you name your design team?"

Why it works: on a design-build or P3 bid the structural engineer is picked while the bidding teams form, often only weeks before award. It tells you how long KOR has to get on the team.

14. "What's your preferred forming system and crane plan?"

Why it works: the contractor's means and methods shape slab layout and repetition. It shows KOR designs for how they build, not just for the drawings.

15. "Which parts are you designing yourselves, through your own engineers or suppliers?"

Why it works: shoring, formwork, connections and pre-engineered items are often designed by the contractor's specialty engineer and sealed by them. It sets the scope line early, before it becomes a dispute.

16. "How fast do you need shop drawings back, and how much notice do you give for site reviews?"

Why it works: these are the two places contractors feel the engineer every week. KOR asks for 24 hours' notice for a field review in normal hours, 7:30 to 4:00 on weekdays, and 48 hours for anything outside them.

17. "On wood, how are you handling the tie-downs as you go up?"

Why it works: KOR's specification assumes rod tie-downs go in floor by floor so the shear walls brace the building during construction. Any other temporary bracing is the contractor's to design and pay for. Asking shows you know the job site.

18. "Where do you usually see the most RFIs and changes on a job like this?"

Why it works: it invites them to tell you what a bad engineer costs them, in their words.

If they say "coordination between consultants": "Let's get our project lead on a call. That's worth talking through before you bid."

QUESTIONS 19 TO 24

About the site and the ground

The ground decides the foundations, and the foundations can swing a budget. The geotechnical engineer owns the soil; the structural engineer owns what sits on it.

19. "Is there a geotechnical report yet, and who's the geotech?"

Why it works: it is the first question on any site. The geotech is usually hired by the owner, sometimes by the design-builder; either way KOR needs to work with them.

If they say "not yet": "Our engineer can tell you what the geotech will need from us: loads, settlement limits and seismic requirements."

20. "Anything soft, wet or filled on the site?"

Why it works: soft or liquefiable soil means piles or ground improvement. In plain words, it signals you know where the foundation money goes.

21. "How close are the neighbours, and how deep are you digging?"

Why it works: shoring and underpinning next to existing buildings involve both geotechnical and structural design, including the extra load the neighbour puts on the soil. Tight urban sites are where this bites.

22. "Is the site sloping, or are there retaining walls?"

Why it works: retaining walls that hold up the building's foundations are structural elements, and a slope changes the foundation design and the excavation plan.

23. "On a tall tower, is anyone planning a site-specific seismic study?"

Why it works: for tall buildings the geotech may provide a site-specific seismic hazard analysis, and EGBC expects the structural and geotechnical engineers to agree who does what. Asking shows you know tall work is a different league.

24. "Are there existing structures on site you want to keep or reuse?"

Why it works: reuse means a condition assessment, which is its own service with its own guideline. It also signals renovation and retrofit work KOR does.

QUESTIONS 25 TO 30

About schedule, cost and risk

These are the questions that turn a technical chat into a business one, and they are the ones decision-makers respond to.

25. "What happens to the project if the structure comes in late or over?"

Why it works: it makes the cost of a weak engineer concrete. When the client answers, they are explaining the value of KOR to themselves.

26. "How much design is done before you go for the building permit?"

Why it works: it shows you know the stages: concept, design development, permit, construction documents, tender and construction. It tells you what scope KOR would be quoting.

27. "Is this going to need a peer review or an independent review?"

Why it works: on tall concrete buildings EGBC requires a documented independent review before drawings go out for construction, and a peer review where advanced analysis is used. Both take time and belong in the schedule.

If they don't know: "That's a question for our engineer. Let me set up a call."

28. "How many site reviews have you budgeted, and how long is construction?"

Why it works: field reviews are a large part of the fee, and KOR's fixed fees cover a reasonable number, not unlimited. Asking early avoids a fight later.

29. "Are you expecting changes after the permit drawings, like unit mix or a sale of the project?"

Why it works: change is the biggest fee risk. KOR bills work outside the agreed scope at hourly rates, so knowing what is likely to change shapes the proposal.

30. "If we could settle the structure early, what would that be worth to you?"

Why it works: a need-payoff question. It gets the client to say the benefit out loud, and gives them words to use when they sell KOR to their partners or board.

HOLD THE LINE

Five things never to say

Each of these feels helpful in the room. Each one either commits KOR to something an engineer hasn't checked, or costs the trust the questions just earned.

✕ NEVER

1. **A number.** No slab thickness, column size, member size or rough steel quantity, "ballpark" included. It will be quoted back to KOR.
2. **A promise on cost or schedule.** Not "we can have permit drawings in eight weeks," not "that'll save you money." Say "Let me find out what's realistic and come back to you."
3. **A word against a competitor.** The client may have hired them, may again, and may repeat it. Talk about what KOR does well.
4. **A code interpretation.** "The code allows that" or "you won't need a peer review" is engineering judgment. It belongs to a registered engineer, and the authority having jurisdiction has the final word.
5. **A commitment of an engineer's time.** Not "John will be at every meeting" or "we'll turn that around tomorrow." Offer the call; let the engineer set the commitment.

WHAT TO SAY INSTEAD

"Good question. I don't want to guess on that. Let me get our engineer on a call this week." Clients hear this as a sign that KOR takes its answers seriously.

ONE PAGE

Cheat sheet: twelve terms from the course

- **Load path:** the route every load takes to the ground: slab, beams, columns or walls, foundations, soil.
- **Lateral system:** whatever keeps the building standing when wind or an earthquake pushes it sideways.
- **Core:** the concrete walls around elevators and stairs; a tower's backbone against earthquakes.

- **Shear wall:** a wall that resists sideways force; in wood frame, the walls that must stack floor to floor.
- **Transfer:** a deep beam or thick slab that carries a column or wall sideways because the support below isn't there. Expensive.
- **PT slab:** a post-tensioned concrete slab, with tensioned steel tendons, so it can be thinner and span further.
- **Part 3 / Part 9:** the two tracks of the BC Building Code: larger or complex buildings that need engineers throughout, and small houses and buildings.
- **SER (Structural Engineer of Record):** the engineer responsible for the structure as a whole and for signing the structural letters of assurance.
- **Letters of Assurance:** the code's schedules where registered professionals commit to design (Schedule B) and confirm field review (Schedule C-B).
- **Field review:** the engineer's site visits during construction to check the work follows the design.
- **Geotech:** the geotechnical engineer, who reports what the soil can carry; the structural engineer designs the foundations on it.
- **Shop drawings:** drawings prepared by fabricators and contractors from KOR's design, which KOR reviews before things are built.

TWO MINUTES

Check yourself

1. **An architect says the core is "still moving around." What do you do?**

   Offer a call with a KOR engineer now. Core size, location and openings are what EGBC says should be agreed early, and moving the core later is expensive.

2. **Why ask a developer what went wrong on their last project?**

   It surfaces real pain, which is the reason they would change engineers. Ask what it cost them before you talk about KOR.

3. **A contractor asks you whether the slab can be thinner. What do you say?**

   "Good question. Let me get our engineer on a call." No numbers, ever.

4. **What is the first question about any site?**

   Is there a geotechnical report, and who is the geotech?

5. **What makes question 30 ("what would that be worth to you?") work?**

The client states the benefit in their own words, and those are the words they will use to sell KOR internally when you are not there.

SOURCES, AND THE REVIEW THIS STILL NEEDS

EGBC's Professional Practice Guidelines for tall concrete buildings, Part 3 buildings, geotechnical services, shop drawings and condition assessments; the joint guidelines for encapsulated mass timber; EGBC's bulletin on five- and six-storey wood frame; the Guide to the Letters of Assurance; KOR's proposal templates, company introductions, specifications and field review procedures; on question technique, *SPIN Selling* and *The Trusted Advisor*. A KOR engineer reads each lesson before it is used with a client.